

IE4-DS LAB SESSION 1 – REPORT

Synchronous Serial Communication with an Audio Codec

Team (group number + team letter as chosen in Moodle, e.g. 02 A)	Team members	
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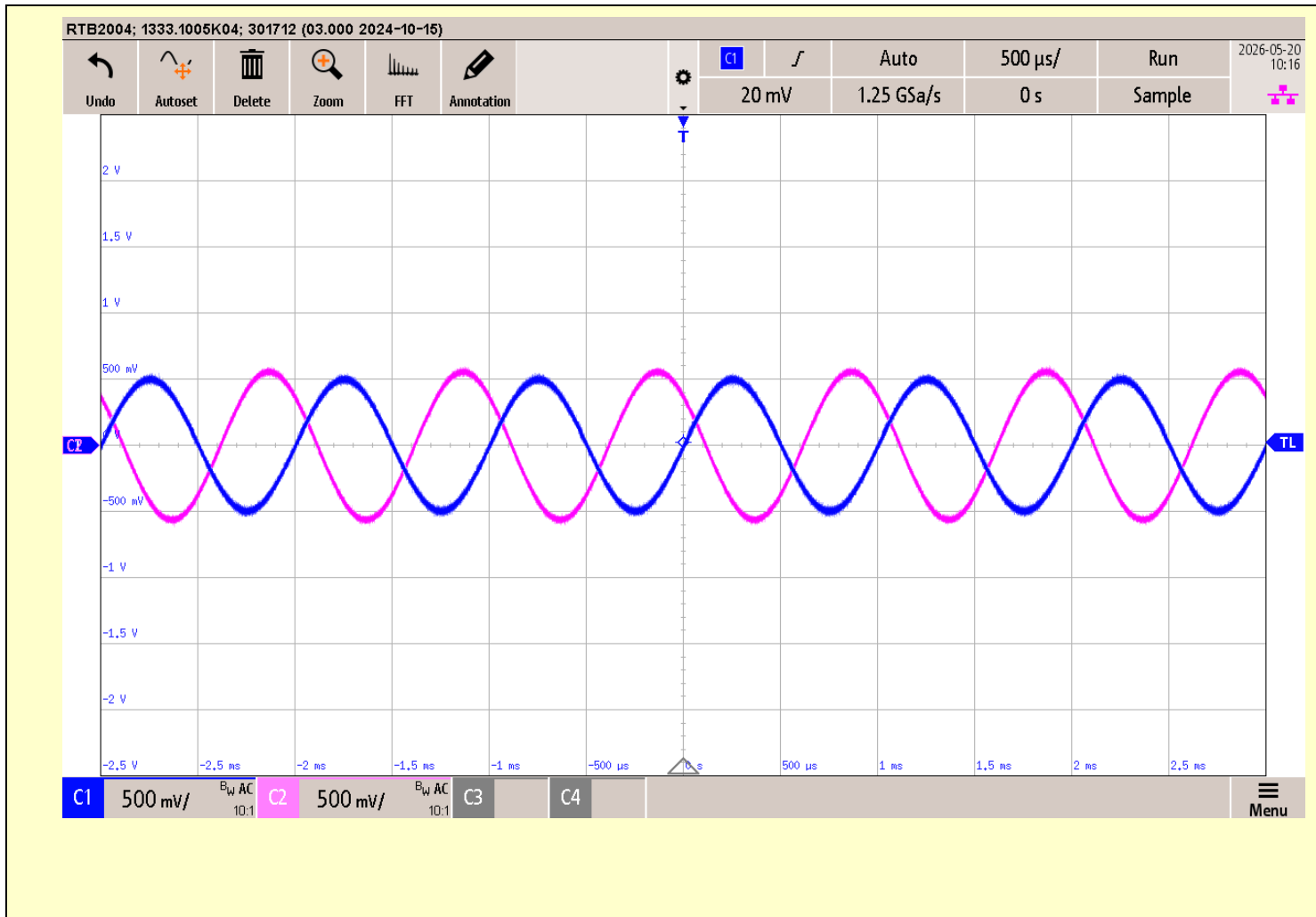
Please fill out the document electronically – including the yellow fields above – and upload it to Moodle once per team at the latest 30 min after the end of the lab session (no matter who uploads it, it will automatically be assigned to all members). Please note that the complete assignment can be found in another document.

Lab Task 1: Test of the system with a sinusoidal signal

With this test you can see if you have implemented everything correctly. The input signal at ... output still does not meet the expectation, continue first with Lab Task 2 for troubleshooting and return later.

- Apply a sinusoidal voltage of e.g. 1 V and 1 kHz to the LINE IN input. Check the amplitude and frequency with the oscilloscope before connecting the function generator. Display both, the input and output signal, on the oscilloscope.
- If the output signal was as expected, slowly turn up the frequency of the sinusoidal

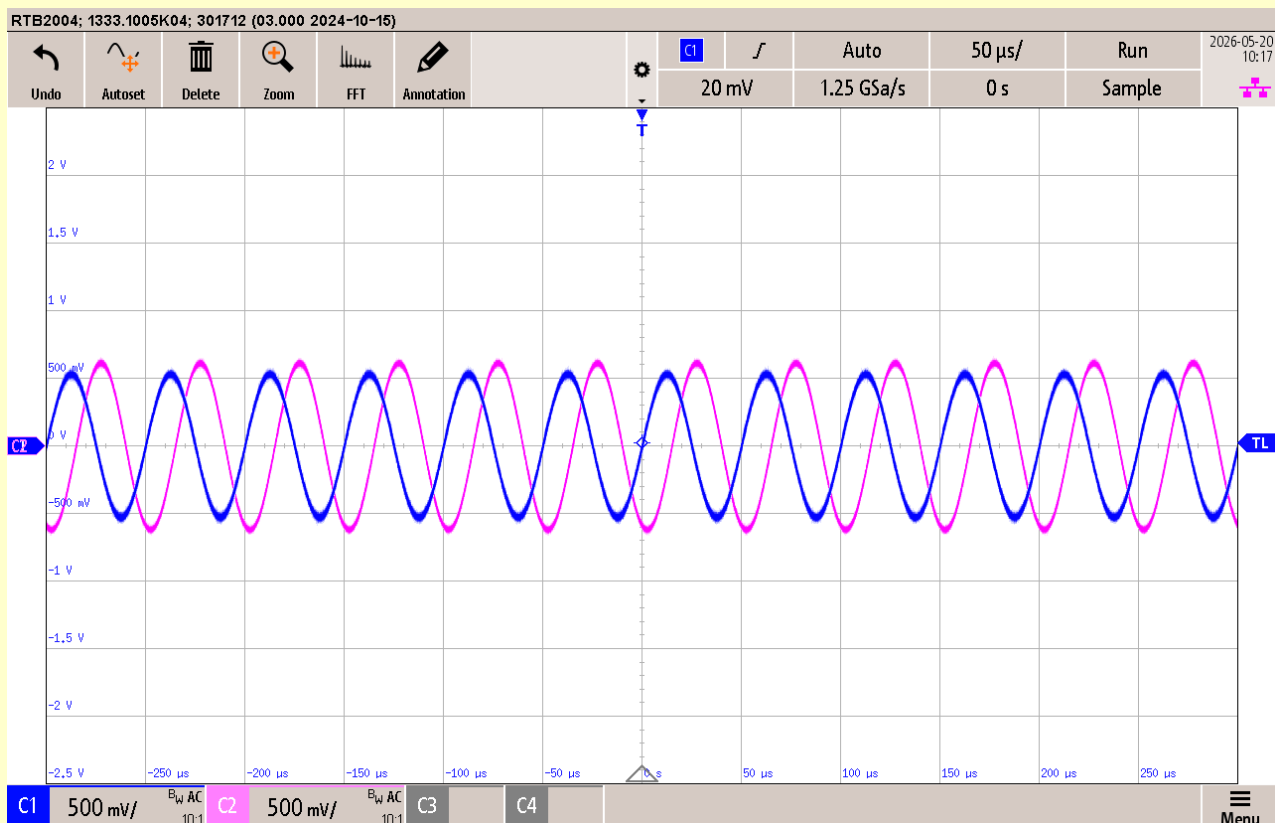
- **Oscilloscope screenshot:**



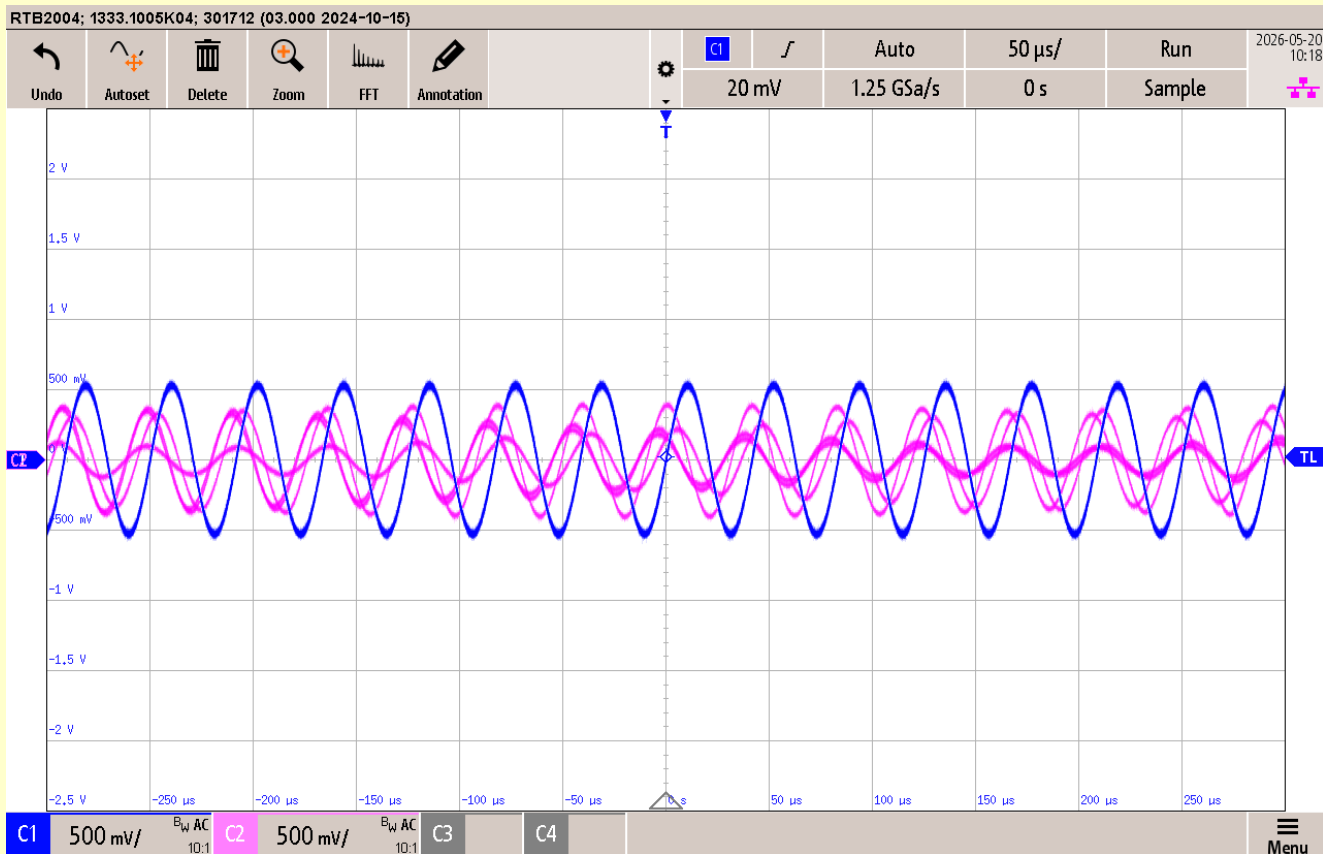
Display both, the input and output signal, on the oscilloscope.

- If the output signal was as expected, slowly turn up the frequency of the sinusoidal signal to 30 kHz. Describe what you observe and name some characteristic frequency and amplitude values.

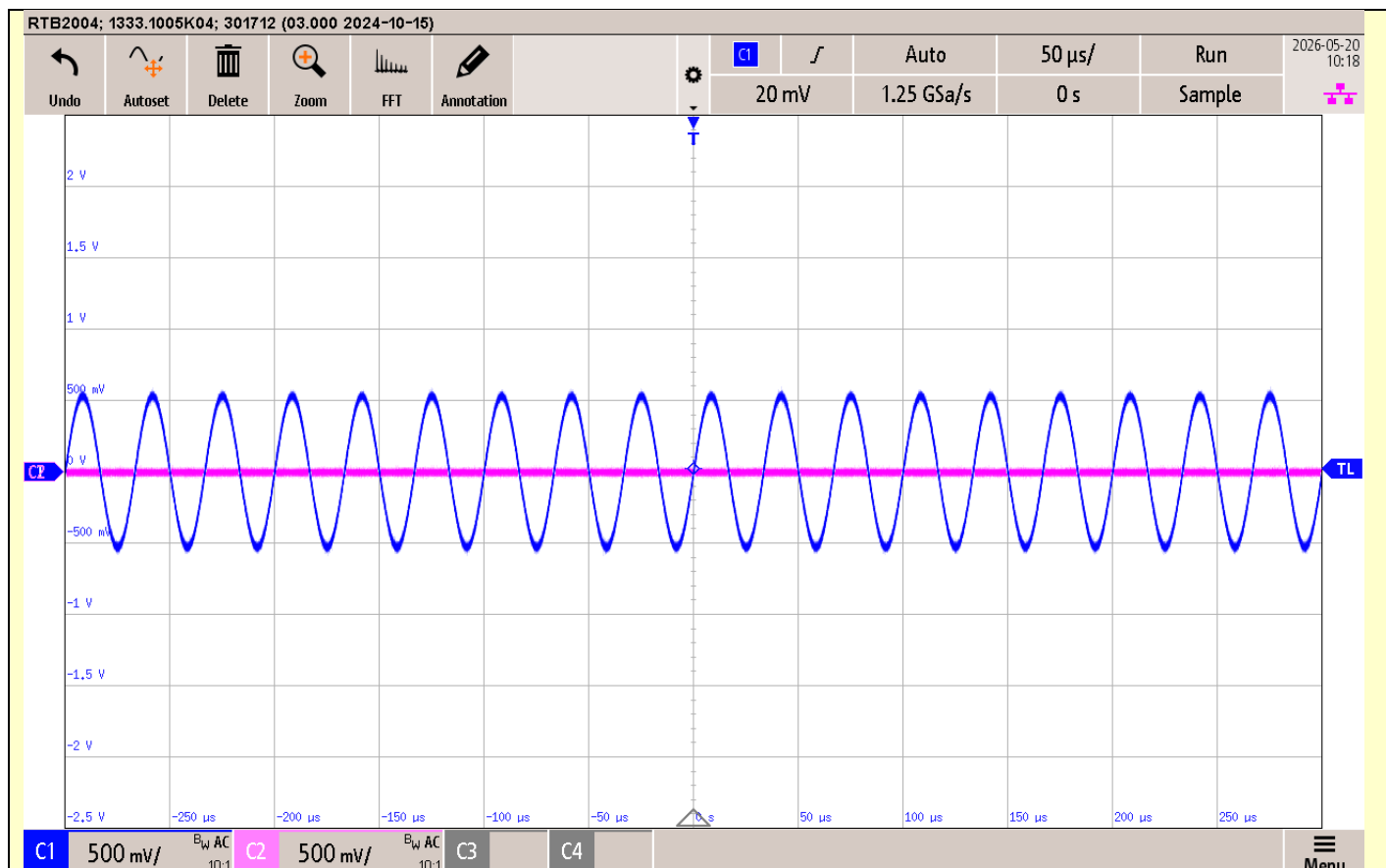
- Oscilloscope screenshots showing the behavior for higher frequencies and verbal description of your observations



20kHz



24kHz



30kHz

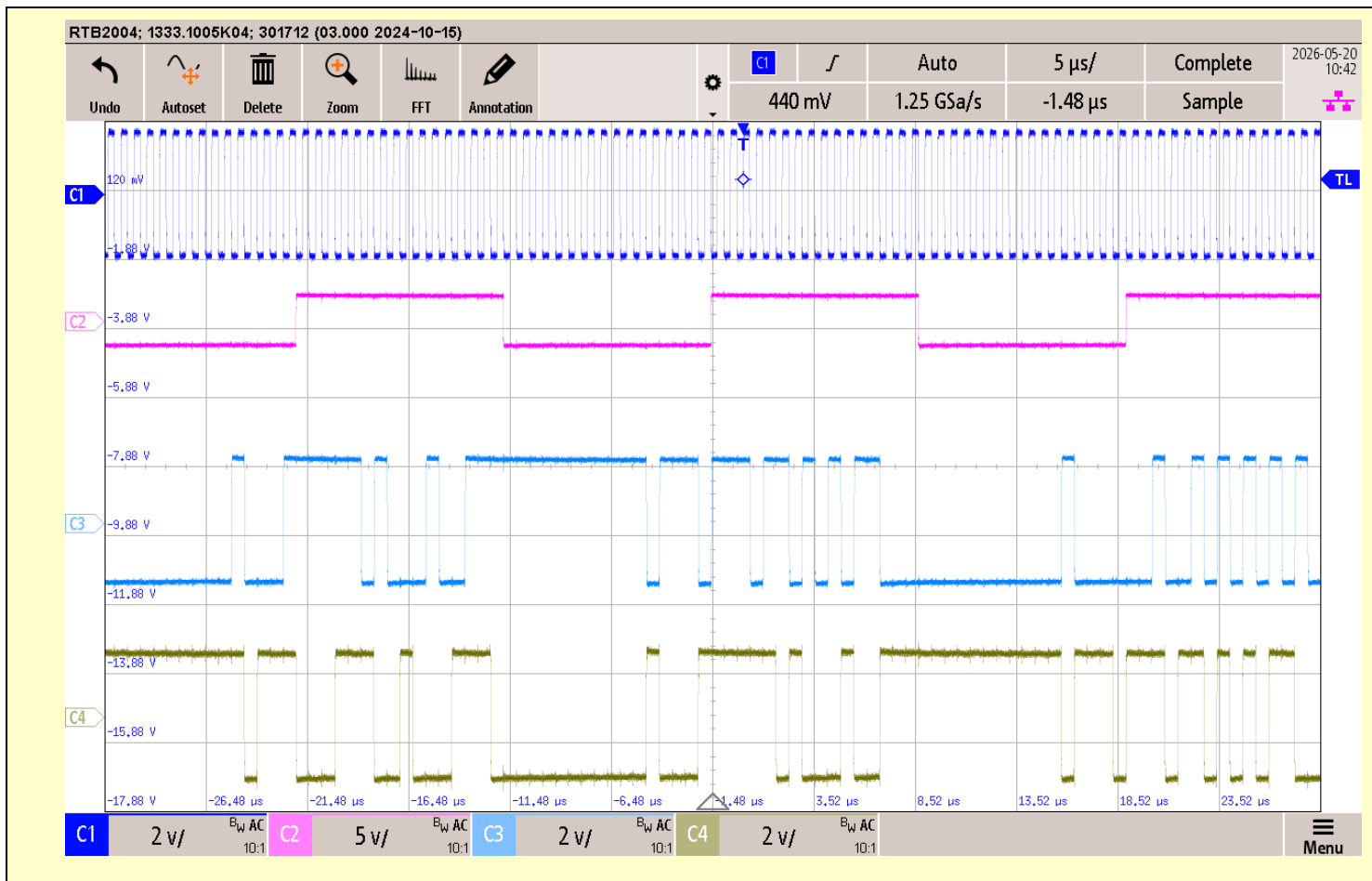
What we observed was that with an increasing frequency, the output signal is being shifted to the right due to the shifting done in our VHDL code. Starting at 24kHz, the output signal's amplitude decreases.

Lab Task 2: Analysis of serial bit streams

In this task, we look at the serial, digital data that is exchanged between the FPGA and the audio codec. If Lab Task 1 did not bring the desired results, here is the possibility to make an

- Apply a sinusoidal voltage of e.g. 1 V and 20 kHz to the LINE IN input. Take a complete data frame of DIN and DOUT together with the BCK and LRC signal. One data frame corresponds to one complete period of the LRC signal with one sample each for left and right channel.

• Oscilloscope screenshot



- Determine the 16 bit sampling values for left and right channel and express them as decimal numbers. Are the values of DIN and DOUT the same? Explain.

• Abtastwerte links / rechts:

Left Channel

DIN – 0010011100100000b – 10,016d

DOUT – 0010011100100000b – 10,016d

Right Channel

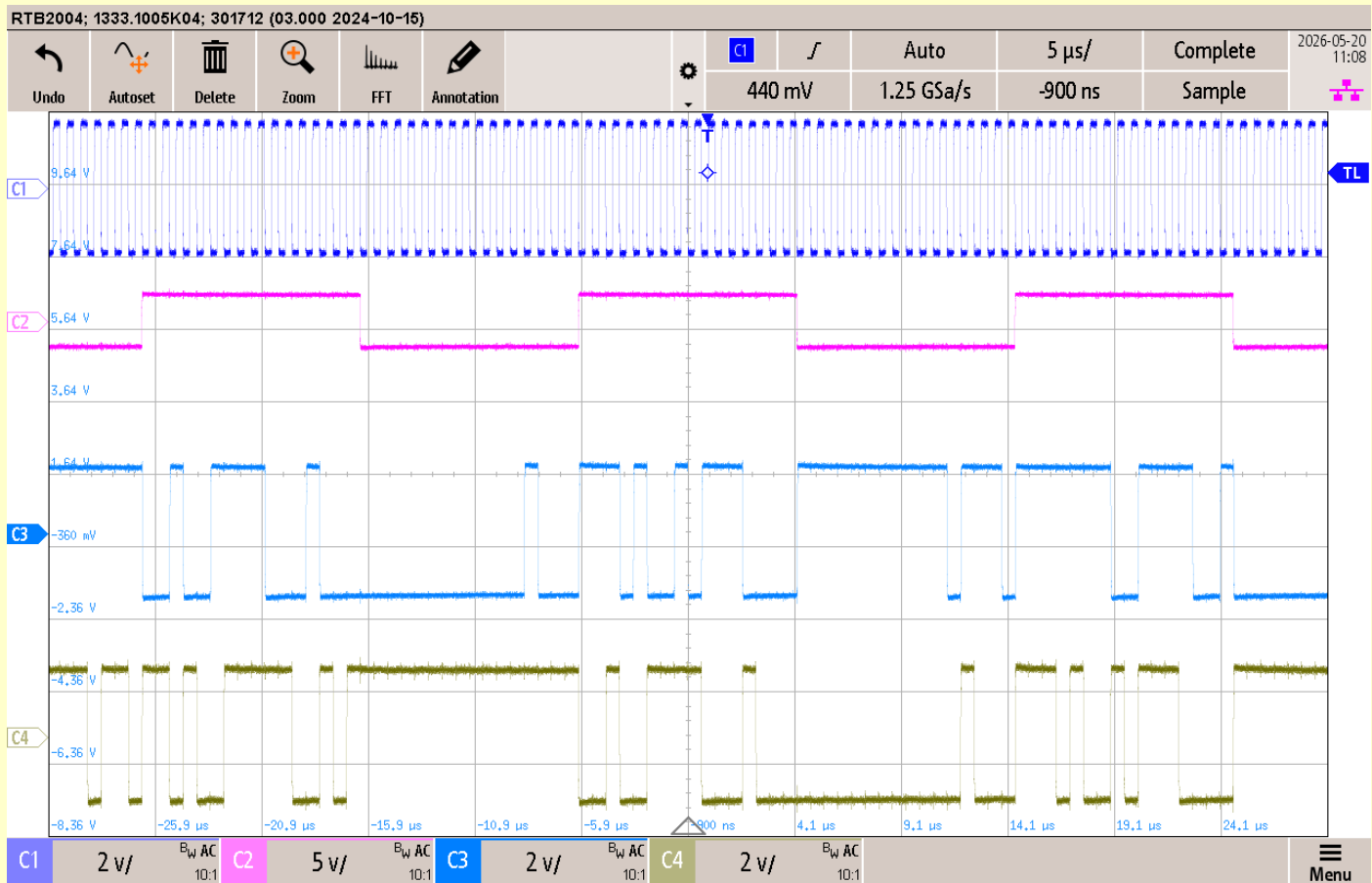
DIN – 1101010010001101 – 54,413d

DOUT – 1101010010001101 - 54,413d

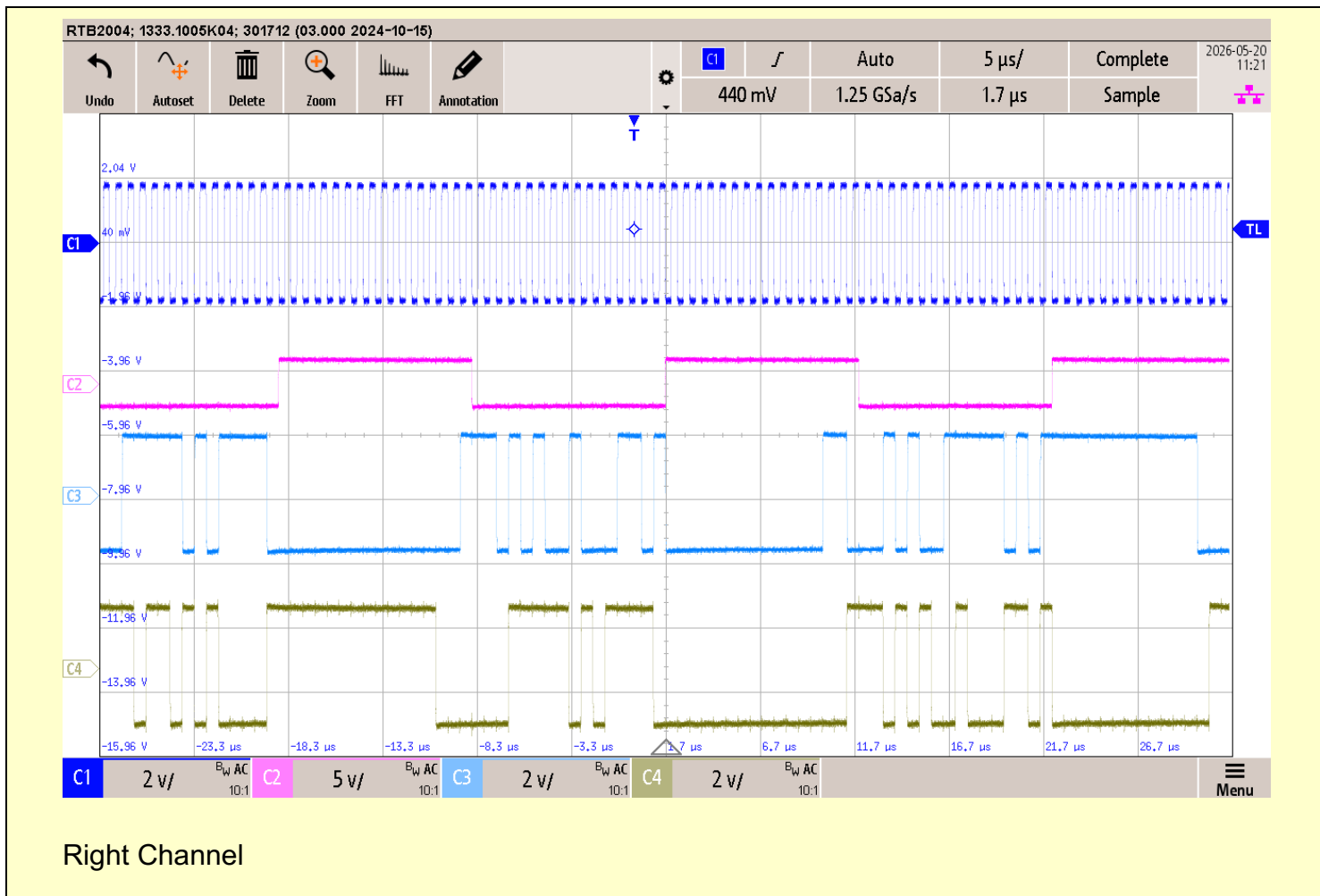
Explanation: The values from both channels are the same because DOUT is simply a time-shifted version of DIN

- Record two consecutive data frames. What can you now state about DIN and DOUT?
- Document the results with oscilloscope images (screenshots, but *not* cell phone photos).

• Oscilloscope screenshot:



Left Channel



• Explanation:

As explained above, DOUT outputs the previous signal of DIN in the next period of LRC.

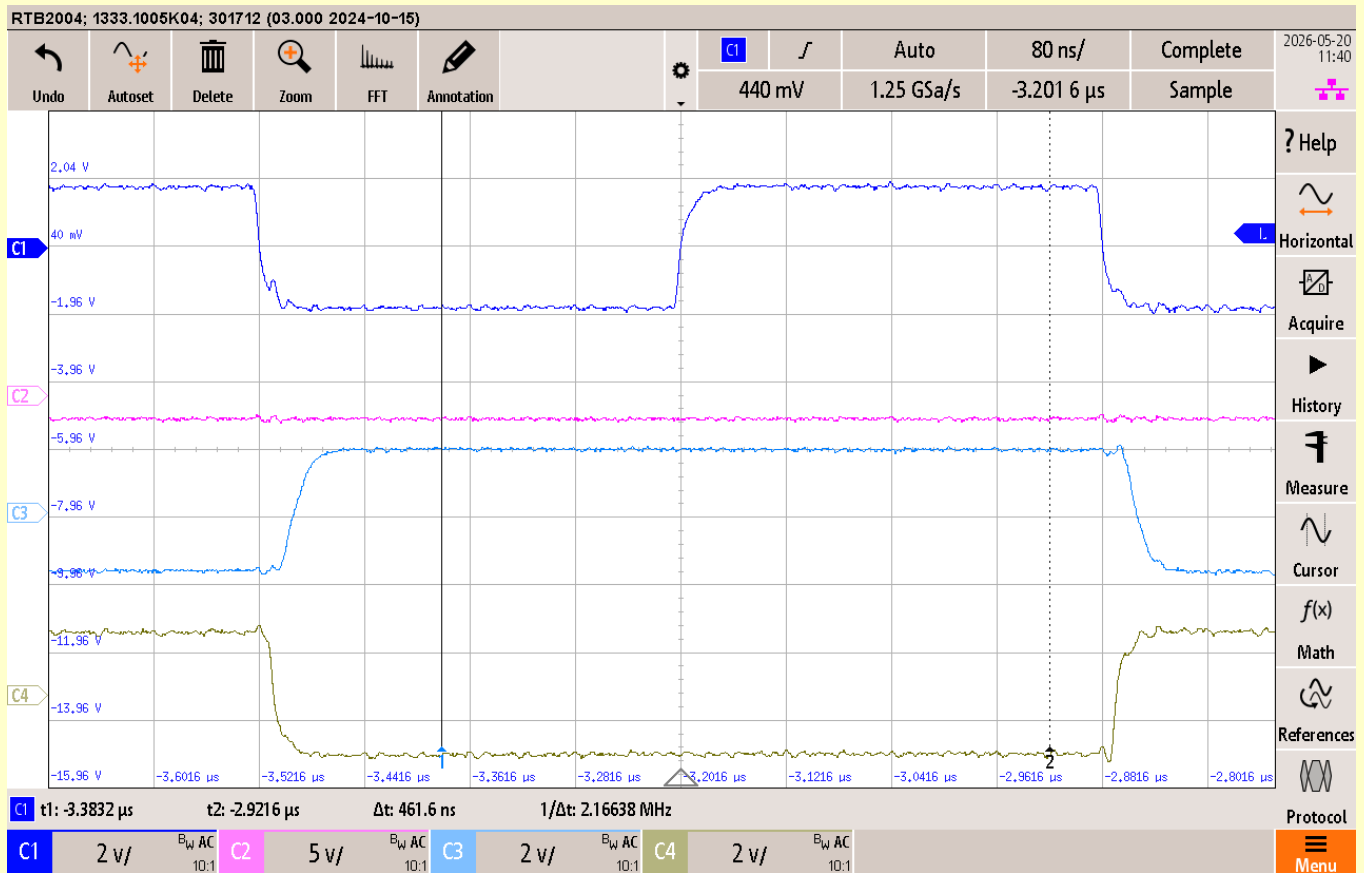
Lab Task 3: Bit Timing

In this part you are to check whether the specifications of the data sheet for the timing behavior of the signals are met. The signal names according to the left column in table 1.3 are used in the following. Measure the following times and compare them with the data

- BCK pulse cycle time $t_{(BCY)}$ (measure at BCK)
- DOUT delay time to BCK falling edge $t_{(BDO)}$ (measure at BCK and DIN)
- DIN setup time $t_{(DIS)}$ (measure at DOUT and BCK)
- DIN hold time $t_{(DIH)}$, (measure at DOUT and BCK)

Only measure two of the four times. You can choose freely, but indicate clearly which ones you have checked for complying with the specifications.

- Oscilloscope screenshots of both measurements



It is possible to measure both $t(\text{BCY})$ and $t(\text{BDO})$ from the following screenshot of the oscilloscope.

- Are the two measured times in accordance with the datasheet specification? Explain.

$t(\text{BCY}) = 635 \text{ ns} \rightarrow \text{DATASHEET: } 300 \text{ ns (min)}$

$t(\text{BDO}) = 24 \text{ ns} \rightarrow \text{DATASHEET: } 40 \text{ ns (max)}$

Explanation: The measurements are in accordance with the datasheet specification because $t(\text{BCY})$ goes over the minimum and $t(\text{BDO})$ does not go over the maximum.

Lab Task 4: Delay of the overall system

First measure the time delay between the analog LINE IN input and the LINE OUT output. Then you are to estimate which part of the delay is due to the audio codec and which part is due to the FPGA.

- What is the risk if you measure the delay with a sinusoidal signal?

- **Explain.**

- Therefore, perform the measurement with a rectangular 1 kHz pulse with a duty cycle of 10% and ± 1 V amplitude. How do you measure the delay you are looking for?
- Explain the shape of the output signal. Is it still rectangular?
- Which part of the delay is due to the audio codec, which due to the FPGA? Use the results of Lab Task 2 for your considerations, too.

- **Oscilloscope screenshots of the measurement**

- **Analyse the delay times and explain**